Simulation of **wildfire** evacuation with dynamic factors and **model** composition

A Beloglazov, M Almashor, E Abebe, J Richter… - **Simulation** Modelling …, 2016 - Elsevier

… **simulation** systems are essential tools for such planning and decision making. During a **wildfire** … Our approach composes several **simulation** and modelling systems, including a **wildfire** …

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Operational **wildfire** suppression modelling: a **review** evaluating development, state of the art and future directions

TJ Duff, KG Tolhurst - International Journal of **Wildland Fire**, 2015 - CSIRO Publishing

… **review**, we focus on the development and use of models that emulate fire suppression for decision support. We **review** … There are two main approaches to dynamic fire **spread simulation**…

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Evaluating a **model** for predicting active crown fire rate of **spread** using **wildfire** observations

ME Alexander, MG Cruz - Canadian Journal of Forest Research, 2006 - cdnsciencepub.com

… With 57 quantitative **wildfire** observations, no study to date … to evaluate a predictive fire **behavior model**. Thus, contrary to … — A **review** of fire weather, fuels, and fire **behavior**. Can. For. …

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DEVS-FIRE: Towards an integrated **simulation** environment for surface **wildfire spread** and containment

L Ntaimo, X Hu, Y Sun - **Simulation**, 2008 - journals.sagepub.com

… applied to an example of fire **spread simulation** [44]. A recent … the **wildfire spread** and containment **simulation model** presented in … We **review** the basics of **wildfire behavior** [56] to set …

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Wildland surface fire **spread** modelling, 1990–2007. 1: Physical and quasi-physical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… readers, and this is the main reason for the inclusion of his work in the current **review**. … Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial …

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Wildland fire spread modelling using cellular automata: evolution in large-scale spatially heterogeneous environments under fire suppression tactics

A Alexandridis, L Russo, D Vakalis… - … of **Wildland Fire**, 2011 - CSIRO Publishing

… We also use the proposed **model** to simulate a much bigger **wildfire** where both evolutionary data for the fire front and the fire-suppression tactics were available. The study concerns the …

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Wildland fire spread modeling using convolutional neural networks

JL Hodges, BY Lattimer - Fire technology, 2019 - Springer

… **spread** requires multiple simulations that may limit the number of scenarios which can be analyzed during an emergency. A new **wildland fire spread model** … **wildland fire spread model** …

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A data-driven **model** for large **wildfire** behaviour **prediction** in Europe

D Rodriguez-Aseretto, D de Rigo, M Di Leo… - Procedia Computer …, 2013 - Elsevier

… as a required input for a realistic **prediction** of the **wildfire dynamics**. In this case, the fire controls are decided elsewhere and notified to the predictive **model**. This may be exploited by …

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A **Survey** of **Wildfire Spread Prediction** and Risk Estimation Methodswith Machine Learning Techniques

S Dascalu, R Wu, A Munoz, S Guan - 2022 - scholarsarchive.byu.edu

… we conducted a **survey** on papers published recently focused on **wildfire spread prediction** and … In the rest of the paper, Section 2 illustrates how **wildfire spread** behaviour is predicted in …

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Modelling smoke transport from wildland fires: a **review**

SL Goodrick, GL Achtemeier, NK Larkin… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… Another grid **model** receiving attention for **wildland fire** related air quality issues is the … of the modelling studies cited in this **review** involves the **dynamics** of the buoyant phase of the …

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